

Knowledge Base (CC BY-SA 4.0)

Machine Learning on Knowledge Bases

Rule Mining Algorithms (I)

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Lecture Contents

1. Rule Mining Algorithm (1)
2. Representation Learning
3. Rule Mining Algorithm (2)
4. Knowledge Acquisition from LLMs
5. TBA

TODAY'S AGENDA

- ▶ What Is Knowledge Base ?
- ▶ Missing Link Completion
- ▶ Horn Rule Mining: AMIE

TODAY'S AGENDA

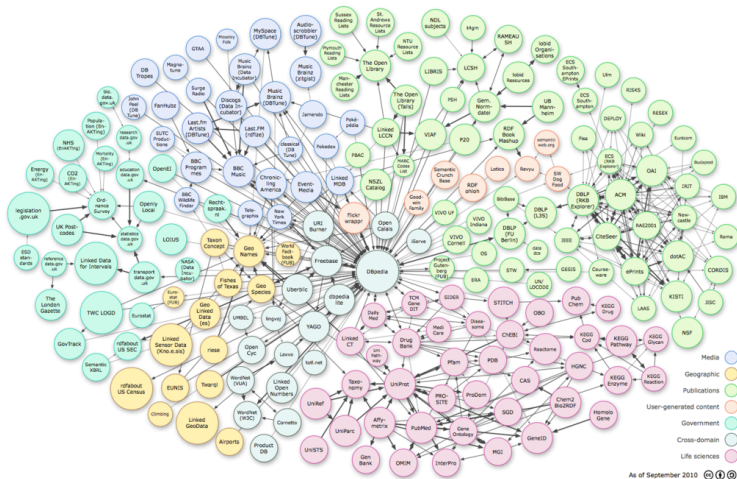
▶ What Is Knowledge Base ?

▶ Missing Link Completion

▶ Horn Rule Mining: AMIE

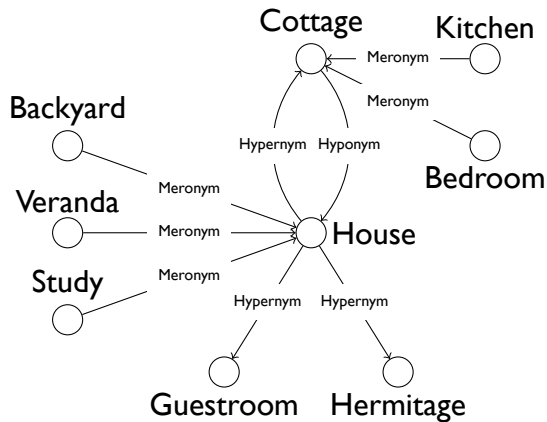
DBPedia (2008)

A project aiming to extract structured content from Wikipedia information



WordNet (1998)

Subgraph related to “House”

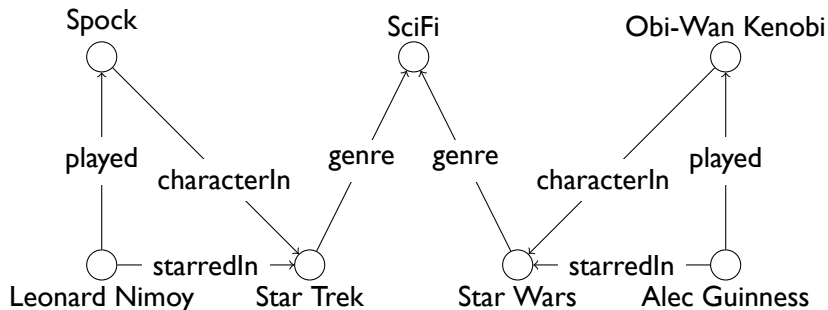


English WordNet :

- ▶ A set of synsets (about 115,000 types)
- ▶ contains about 150,000 words
- ▶ Relationship between synsets
 - ▶ hypernym • hyponym • holonym • meronym

Manually constructed

Definition: RDF Knowledge Base



KB $\mathcal{K}$ is a set of facts

- **fact:** a triple of the form (s, r, o) (alternative notation $r(s, o)$)
- s/o is the **subject/object**, r is the **relation** (or predicate)

Open Knowledge Bases

The research on Semantic Web and Linked Data led to many open datasets

These open datasets are rebranded as “Knowledge Graphs”

- ▶ DBPedia, Freebase, YAGO, NELL, Wikidata, KBPedia, Datacommons.org

Many open Knowledge Bases are sourced from Wikipedia and also benefited from unstructured corpus for their building process

Problem: KBs are incomplete and have many missing links

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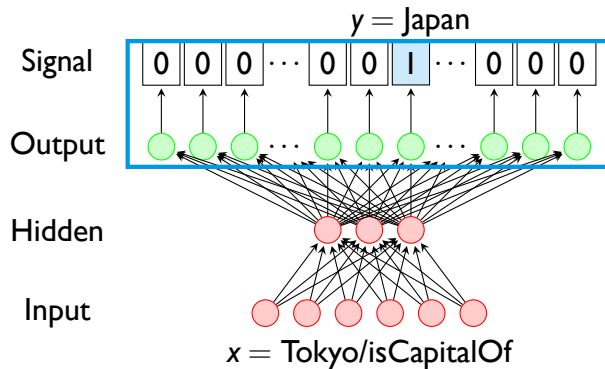
Missing Link Prediction

Missing link prediction in the context of KBs is also referred to as knowledge graph completion

Statistical Relational Learning in KBs

- ▶ (Deep) Representation Learning Model
- ▶ Feature-based Regression Model
- ▶ **Rule-based Logical Inference Model**

Representation Learning in KBs



Learning : minimize errors between signal and output

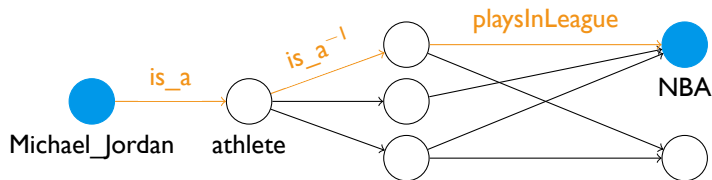
Loss Function :
e.g. Cross-Entropy Loss

$$-\log \frac{\exp(f_{\theta}(x, y))}{\sum_{y' \in \mathcal{Y}} \exp(f_{\theta}(x, y'))}$$

“Knowledge Graph Embedding: A Survey of Approaches and Applications”, Wang et. al., IEEE TKDE, 2017.

Feature-based Learning in KBs

Relational Path: $\mathbf{r} = r_{k_1} / r_{k_2} / \dots / r_{k_n}$



Path Ranking: predict a relation r_k between e_i and e_j using path features

$$s_{ij}^{(r_k)} = \sum_{\mathbf{r} \in \mathbf{P}} \theta_{\mathbf{r}}^{(r_k)} f_{\mathbf{r}}(e_i, e_j)$$

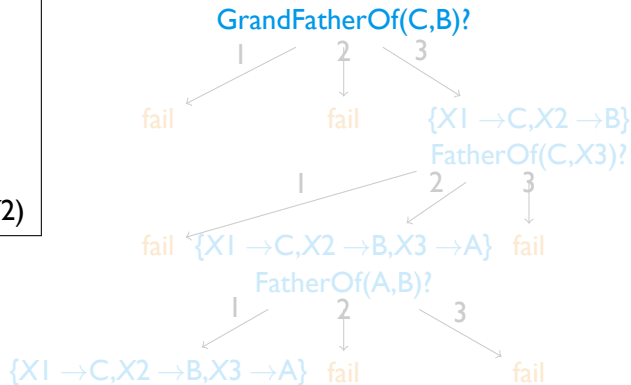
- ▶ $\mathbf{P}$: a set of relational paths
- ▶ $f_{\mathbf{r}}(e_i, e_j)$: a path feature
- ▶ $\theta_{\mathbf{r}}^{(r_k)}$: a weight of a path $\mathbf{r}$ for predicting r_k

“Random Walk Inference and Learning in A Large Scale Knowledge Base”, Lao et. al., EMNLP, 2011.

Logical Deductive Inference

KB Example:

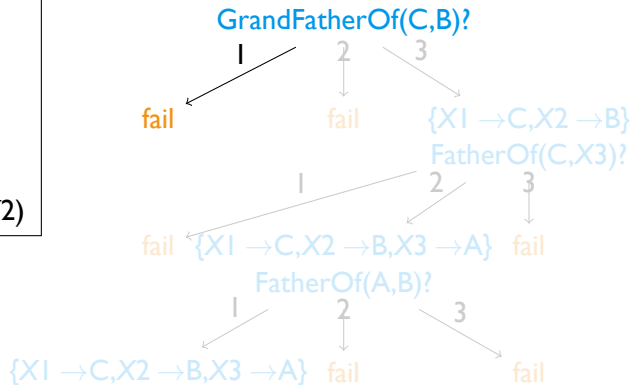
1. $\text{FatherOf}(A,B)$
2. $\text{FatherOf}(C,A)$
3. $\text{FatherOf}(X1,X3) \wedge \text{FatherOf}(X3,X2) \Rightarrow \text{GrandFatherOf}(X1,X2)$



Logical Deductive Inference

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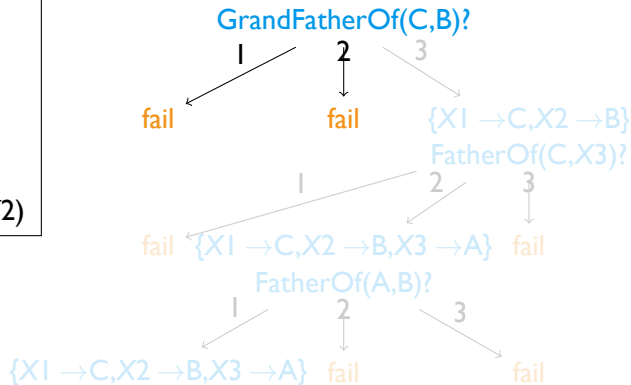
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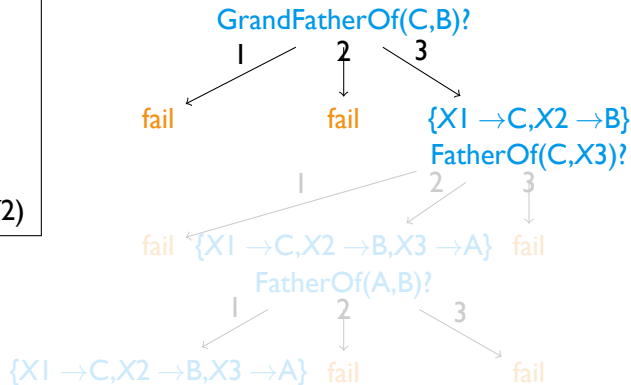
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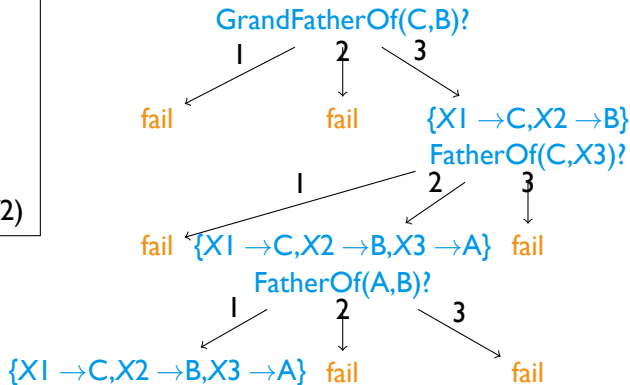
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- ▶ Missing Link Completion
- ▶ Horn Rule Mining: AMIE

AMIE: Association Rule Mining

Horn Rule Mining:

- ▶ “AMIE: Association Rule Mining under Incomplete Evidence in Ontological Knowledge Bases”, Galárraga et. al., WWW, 2013.
- ▶ “Fast Rule Mining in Ontological Knowledge Bases with AMIE+”, Galárraga et. al., VLDB, 2015.

AMIE is faster than classical ILP (Inductive Logic Programming) methods and consistent with Open World Assumption (OWA) in KGs

- ▶ OWA: a fact that is not contained in the KB is not necessarily false

Atoms and Horn Rules

Atom: a fact that can have variables at the subject and/or object position

Rule Example:

$$hasChild(X1, X2) \wedge isCitizenOf(X1, X3) \Rightarrow isCitizenOf(X2, X3)$$

Horn Rule: a rule with Body and Head by implication

$$\underbrace{B_1 \wedge B_2 \wedge \cdots \wedge B_n}_{\text{Body}} \Rightarrow \underbrace{r(X, Y)}_{\text{Head}}$$

where Body is a set of Atoms and Head is a single atom (Abbreviation $\vec{B} \Rightarrow r(X, Y)$)

Rule Restrictions

AMIE searches only **connected** and **closed** rules

Connected:

- ▶ Two Atoms in a rule are connected: They share a variable or an entity
 - ▶ e.g.: $r_1(x,y)$ and $r_2(y,z)$
- ▶ Connected rule: Every atom is transitively connected to every other atom
 - ▶ e.g.: $r_1(x,y) \wedge r_2(y,z) \wedge r_3(x,w) \Rightarrow r(x,y)$

Closed:

- ▶ A variable in a rule is closed if it appears at least twice in the rule
- ▶ A rule is closed if all its variables are closed

Instantiation and Prediction

Instantiation of a rule: a copy of the rule where all variables have been substituted by constants

- ▶ e.g.: An instantiation of a rule $\text{livesIn}(x, y) \Rightarrow \text{wasBornIn}(x, y)$ is $\text{livesIn}(\text{Adam}, \text{Paris}) \Rightarrow \text{wasBornIn}(\text{Adam}, \text{Paris})$

Prediction of a rule: the head atom of an instantiated rule if all body atoms of the instantiated rule appear in the KB $\mathcal{K}$

- ▶ Notation: $\mathcal{K} \wedge R \models p$
- ▶ e.g.: The prediction of the instantiated rule $\text{livesIn}(\text{Adam}, \text{Paris}) \Rightarrow \text{wasBornIn}(\text{Adam}, \text{Paris})$ is the head atom $\text{wasBornIn}(\text{Adam}, \text{Paris})$ ($\text{livesIn}(\text{Adam}, \text{Paris}) \in \mathcal{K}$)

The Aim of Rule Mining

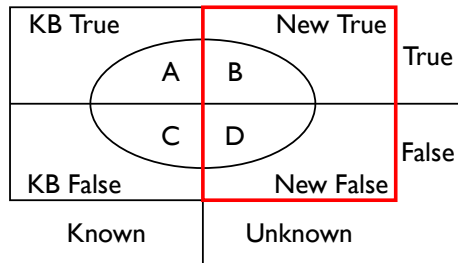
KB True	New True	True
A	B	Predictions
KB False	New False	
C	D	False
Known	Unknown	

1. KB True: True facts that are known to the KB
2. New True: True facts that unknown to the KB
3. KB False: Facts that are known to be false in the KB
4. New False: Facts that are false but unknown to the KB

The aim of rule mining is to find rules that make true predictions

- ▶ maximize the area B and minimize the area D
- ▶ B and D are unknown and we need to design good measures of rules

The Aim of Rule Mining



1. KB True: True facts that are known to the KB
2. New True: True facts that unknown to the KB
3. KB False: Facts that are known to be false in the KB
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The aim of rule mining is to find rules that make true predictions

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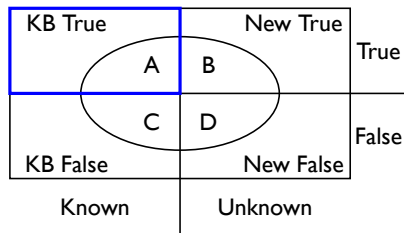
Support and Head-coverage

Support:

$$\text{support}(R) := |\{p : (\mathcal{K} \wedge R \models p) \wedge p \in \mathcal{K}\}|$$

Head-coverage:

$$\begin{aligned} & hc(\vec{B} \Rightarrow r(x, y)) \\ &:= \frac{\text{support}(\vec{B} \Rightarrow r(x, y))}{|\{(x, y) : r(x, y) \in \mathcal{K}\}|} \\ &= \frac{\text{support}(\vec{B} \Rightarrow r(x, y))}{\text{size}(r(x, y))} \end{aligned}$$



The support of a rule quantifies only the number of known correct predictions of the rule

KB Example $\mathcal{K}$:

<i>livesIn</i>	<i>wasBornIn</i>
(Adam, Paris)	(Adam, Paris)
(Adam, Rome)	(Carl, Rome)
(Bob, Zurich)	

Rule Example R :

$$\text{livesIn}(x, y) \Rightarrow \text{wasBornIn}(x, y)$$

Support:

$$\text{support}(R) = |\{p : (\mathcal{K} \wedge R \models p) \wedge p \in \mathcal{K}\}| = 1$$

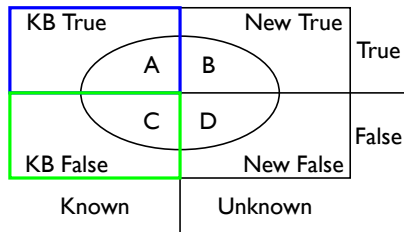
When $x = \text{Adam}$ and $y = \text{Paris}$, the prediction of R is $\text{wasBornIn}(\text{Adam}, \text{Paris}) \in \mathcal{K}$

Head-coverage:

$$\text{hc}(R) = \frac{\text{support}(R)}{|\{(x, y) : r(x, y) \in \mathcal{K}\}|} = 1/2$$

PCA and Confidence

The central challenge is to provide counter-examples (negative samples) for the rule mining



Confidence: $cex(R)$ is a set of counter examples for a rule R

$$conf(R) := \frac{support(R)}{support(R) + \underbrace{|\{p : (\mathcal{K} \wedge R \models p) \wedge p \in cex(R)\}|}_{\text{The number of false predictions}}}$$

Partial Completeness Assumption (PCA): whenever at least one object for a given subject and a relation is in a KB, then all objects for that subject-relation pair are assumed to be known

- ▶ PCA relies on the fact that relations in KBs tend to be *functional*
- ▶ e.g.: The relation *hasBirthday*(x, y) is
function (*hasBirthday*(Russell, 18_May_1872))

PCA confidence:

$$\begin{aligned} \text{conf}_{pca}(R) &= \frac{\text{support}(R)}{\text{support}(R) + |(x, y) : (\mathcal{K} \wedge R \models r(x, y)) \wedge r(x, y') \in \mathcal{K} \wedge r(x, y) \notin \mathcal{K}|} \\ &= \frac{\text{support}(R)}{|(x, y) : (\mathcal{K} \wedge R \models r(x, y)) \wedge r(x, y') \in \mathcal{K}|} \end{aligned}$$

KB Example $\mathcal{K}$:

<i>livesIn</i>	<i>wasBornIn</i>
(Adam, Paris)	(Adam, Paris)
(Adam, Rome)	(Carl, Rome)
(Bob, Zurich)	

Rule Example R :

$livesIn(x, y) \Rightarrow wasBornIn(x, y)$

PCA Confidence:

$$\begin{aligned} Conf_{pca}(R) &= \frac{support(R)}{|(x, y) : (\mathcal{K} \wedge R \models r(x, y)) \wedge r(x, y') \in \mathcal{K}|} \\ &= 1/2 \end{aligned}$$

1. the prediction $wasBornIn(Adam, Paris)$ is a positive example
2. the prediction $wasBornIn(Adam, Rome)$ is a negative example because we already know a different place of birth for Adam

AMIE Algorithm

```
1: function AMIE(KB  $\mathcal{K}$ ,  $minHC$ ,  $maxLen$ ,  $minConf$ )
2:    $q := [r_1(x, y), r_2(x, y), \dots, r_m(x, y)] \Leftarrow$  Initialize head atoms
3:    $out := \langle \rangle \Leftarrow$  Initialize an output list
4:   while  $q$  is not empty do
5:      $r = q.dequeue()$ 
6:     // Decide if a rule  $r$  should be output or not
7:     if AcceptedForOutput( $r, out, minConf$ ) then
8:        $out.add(r)$ 
9:       if  $length(r) < maxLen$  then
10:         $R(r) = Refine(r) \Leftarrow$  Add a new atom to the body of  $r$ 
11:        for all rules  $r_c \in R(r)$  do
12:          if  $hc(r_c) \geq minHC \wedge r_c \notin q$  then
13:             $q.enqueue(r_c)$ 
14:   return  $out$ 
```

Summary

Missing Link Completion

- ▶ Logical Inference-based Model
- ▶ **Horn Rule Mining Algorithm**

For the details:

- ▶ “Fast Rule Mining in Ontological Knowledge Bases with AMIE+”, Galárraga et. al., VLDB, 2015.
- ▶ source code: <https://github.com/lajus/amie>

Homework

Question: Calculate Support and PCA Confidence scores for the following rule.

Rule: $livesIn(x, y) \Rightarrow isPoliticianOf(x, y)$

KB:

<i>livesIn</i>	<i>isPoliticianOf</i>	<i>diedIn</i>
(Ava, Paris)	(Ava, Paris)	(Ava, Paris)
(Emily, London)	(Bob, Newyork)	(Bob, Newyork)
(Emily, Paris)	(Ava, Newyork)	(Emma, Lisbon)

Please be sure to submit your report to Classroom by the next class